

Collection API Assignments

Assignment 1: Student Attendance Tracker (List)

Topic: ArrayList, iteration, and basic list operations

Level: Beginner

Problem Statement:

You are developing a system for a school to track daily attendance.

- Maintain a list of student names who attended class on a particular day.
- Allow the user to:
 - Add a student's name to the attendance list
 - Remove a student's name if added by mistake
 - Display the final attendance list in the order students were marked present
 - Check if a particular student attended the class

Assignment 2: Unique City Names for Delivery Service (Set)

Topic: HashSet, uniqueness, searching

Level: Beginner

Problem Statement:

You're building a logistics system for a delivery service.

- As the company expands, it starts delivering to new cities.
- Implement a feature to track all **unique city names** where deliveries have been made.
- The system should allow adding city names, prevent duplicates, and support:
 - Displaying all cities
 - Checking if delivery is available in a particular city

Assignment 3: Employee Directory (Map)

Topic: HashMap, key-value pairs

Level: Intermediate

Problem Statement:

Create an internal directory for a company's HR department.

- Store employee information using employee ID as key and name as value.
- Allow operations to:
 - Add a new employee
 - Remove an employee by ID
 - Fetch an employee's name by ID
 - Display all employee IDs and names

Assignment 4: Product Stock Management (Map + List)

Topic: Map<String, List<String>>, nesting collections

Level: Intermediate

Problem Statement:

A supermarket maintains a stock list categorized by product types.

- For example, "Fruits" → ["Apple", "Banana"], "Beverages" → ["Juice", "Soda"]
- Build a system that can:
 - Add a new product under a category
 - Remove a product from a category
 - List all products under a given category
 - Display the entire stock grouped by categories

Assignment 5: Bookstore Catalog (TreeMap & TreeSet)

Topic: TreeMap, TreeSet, sorting

Level: Intermediate

Problem Statement:

You're building a catalog system for a bookstore.

- Books are categorized by genres. Each genre should be sorted alphabetically.
- Within each genre, books should also be stored in a sorted order.
- Implement functionality to:
 - Add a book to a genre
 - Remove a book from a genre
 - Display all genres and their books in sorted order
 - List all books under a specific genre